We can test the analyzer using the following code snippet:

```
POST index_with_custom_analyzer/_analyze
{
  "text": "<H1>HELLO, WoRLD</H1>",
  "analyzer": "custom_analyzer"
}
```

This program produces two tokens: `["HELLO", "WORLD"]`. Our `html_strip` filter removed the `H1` HTML tags before letting the `standard` tokenizer split the field into two tokens based on a whitespace delimiter. Finally, the tokens were uppercased as they passed through the `uppercase` token filter.

Customization helps satisfy a range of requirements, but we can achieve even more advanced results. Let's take a look.

7.4.1 Advanced customization

While the default configurations of the analyzer components work most of the time, sometimes we need to create analyzers with nondefault configurations for their components. For example, we may want to use a `mapping` character filter to map the character `&` to the word *and* and the characters `<` and `>` to *less than* and *greater than*, respectively.

Suppose our requirement is to develop a custom analyzer that parses text for Greek letters and produces a list of Greek letters. The following code creates an index with these analysis settings.

Listing 7.25 Extracting Greek letters with a custom analyzer

```
PUT index_with_parse_greek_letters_custom_analyzer
{
  "settings": {
    "analysis": {
      "analyzer": {
        "greek_letter_custom_analyzer":{
          "type":"custom",
          "char_filter":["greek_symbol_mapper"],
          "tokenizer":"standard",
          "filter":["lowercase", "greek_keep_words"]
        }
      },
      "char_filter": {
        "greek_symbol_mapper":{
          "type":"mapping",
          "mappings":[
            "α => alpha",
            "β => Beta",
            "γ => Gamma"
          ]
        }
      },
      "filter": {
        "greek_keep_words":{
```

Creates a custom Greek-letter-parser analyzer

The custom analyzer is made of a custom char_filter.

A standard tokenizer tokenizes the text.

Supplies two token filters; greek_keep_words is defined next.

Defines the Greek letters and maps them to English words

The actual mappings: a list of symbols and corresponding values

We don't want to index all the field values, only words that match the keep words.